

Name: _____

Date: _____

DIVISION, (LONG DIVISION)

LO: I can divide larger numbers using the long division algorithm (bus stop method).

What is long division?

Long division is a written method for dividing by **2-digit or larger** numbers. We work digit-by-digit from left to right.

Bus stop: the divisor sits outside, the dividend underneath, and the answer appears on top. Subtract, bring down, repeat.

Steps to remember

●	Divide, multiply, subtract, bring down	DMSB cycle
●	$672 \div 12 = 56$	no remainder
●	$415 \div 13 = 31 \text{ r } 12$	with remainder
●	Skip a step in DMSB	leads to errors
●	Forget place value alignment	answer wrong

Key idea

Repeat the cycle: **D**ivide, **M**ultiply, **S**ubtract, **B**ring down.
Stop when no more digits to bring down.



Repeat until done

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - no remainder

Calculate $672 \div 12$

12 into 6? No. 12 into 67? 5
($5 \times 12 = 60$)

$67 - 60 = 7$. Bring down 2 72

12 into 72? 6 ($6 \times 12 = 72$)

= 56

Answer: 56 (no remainder). Check: $56 \times 12 = 672$

Example 2 - with remainder

Calculate $415 \div 13$

13 into 41? 3 ($3 \times 13 = 39$)

$41 - 39 = 2$. Bring down 5 25

13 into 25? 1 ($1 \times 13 = 13$). $25 - 13 = 12$

= 31 r 12

31 remainder 12. Check: $31 \times 13 + 12 = 403 + 12 = 415$

Example 3 - 3-digit divisor

Calculate $4500 \div 25$

25 into 45? 1 ($1 \times 25 = 25$). $45 - 25 = 20$

Bring down 0 200. 25 into 200? 8

$200 - 200 = 0$. Bring down 0. 25 into

0? 0

= 180

Check: $180 \times 25 = 4500$

Example 4 - decimal answer

Calculate $35 \div 4$

4 into 35? 8 ($8 \times 4 = 32$). $35 - 32 = 3$

Add a decimal point and zero: 30

4 into 30? 7 ($7 \times 4 = 28$). $30 - 28 = 2$ —

continue

= 8.75

Keep going past decimal point until remainder is 0 or you have enough d.p.

YOUR TURN – PRACTICE (deliberate practice)

1. Single-digit divisor

Use the bus-stop method.

- a) $144 \div 6$
- b) $196 \div 7$
- c) $315 \div 9$
- d) $568 \div 8$

2. Two-digit divisor (no remainder)

Divide carefully — line up digits.

- a) $672 \div 12$
- b) $936 \div 13$
- c) $1056 \div 16$
- d) $1638 \div 18$

3. With remainder

Write the answer as 'quotient r remainder'.

- a) $415 \div 13$
- b) $500 \div 17$
- c) $847 \div 23$
- d) $1234 \div 41$

4. Decimal answers

Continue past the decimal point until exact.

- a) $35 \div 4$
- b) $63 \div 8$
- c) $27 \div 5$
- d) $100 \div 8$

5. Mixed long division

Choose the right method.

- a) $504 \div 14$
- b) $725 \div 25$
- c) $847 \div 11$
- d) $1000 \div 8$
- e) $234 \div 18$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Always divide left to right ☐

Correct answer: _____

Reason: _____

b) DMSB stands for Divide, Multiply, Subtract, Bring down ☐

Correct answer: _____

Reason: _____

c) If divisor doesn't fit, skip the digit ☐

Correct answer: _____

Reason: _____

d) Remainder must be smaller than divisor ☐

Correct answer: _____

Reason: _____

e) $100 \div 8$ has remainder 4 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A factory has **2160 chocolates** to pack into boxes of **24**. How many full boxes? Are any chocolates left over? Show your full long division.

Expression: _____

Simplified: _____

b) Mr Khan drives **462 miles** using **14 gallons** of fuel. Find his miles-per-gallon (mpg). Now do **637 miles $\div$ 13 gallons**. Which trip was more efficient?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

